PROGRAMMING IN JAVA

CSA0915

ASSIGNMENT-CO1

CAMPUS SMART-PARKING SYSTEM

TASK 06 – Inheritance Hierarchy

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1. Task06.md

Aim

To demonstrate inheritance in Java by creating a vehicle hierarchy for the Campus Smart-Parking System.

Inheritance

Inheritance allows one class to acquire the properties and methods of another class using the `extends` keyword.

Classes Used

1. **Vehicle:** Parent class containing vehicle number and display method.
2. **Car:** Child class representing cars.
3. **TwoWheeler:** Child class representing two-wheelers.
4. **EV:** Child class representing electric vehicles.

Advantages

- Code reusability
- Easy maintenance
- Reduces code duplication
- Supports method overriding and polymorphism

Conclusion

Inheritance creates a clear relationship between vehicle types and allows common vehicle details to be reused in the parking system.

2. Vehicle.java

```
public class Vehicle {
        String vehicleNumber;
        Vehicle(String vehicleNumber) {
           this.vehicleNumber = vehicleNumber;
        }
        void displayDetails() {
           System.out.println("Vehicle Number: " + vehicleNumber);
        }
    }
```

3. Car.java

```
        public class Car extends Vehicle {
           Car(String vehicleNumber) {
              super(vehicleNumber);
           }
           void showType() {
              System.out.println("Vehicle Type: Car");
           }
        }
```

4. TwoWheeler.java

```
public class TwoWheeler extends Vehicle {
   TwoWheeler(String vehicleNumber) {
      super(vehicleNumber);
   }
   void showType() {
      System.out.println("Vehicle Type: Two-Wheeler");
   }
}
```

5. EV.java

```
        public class EV extends Car {
           EV(String vehicleNumber) {
```

```
        super(vehicleNumber);
    }
    void showType() {
        System.out.println("Vehicle Type: Electric Vehicle");
    }
}
```

6. Main.java

```
public class Main {
    public static void main(String[] args) {
        Car car = new Car("TN01AB1234");
        TwoWheeler bike = new TwoWheeler("TN02CD5678");
        EV ev = new EV("TN03EF9012");
        System.out.println("Campus Parking Vehicle Details:");
        car.displayDetails();
        car.showType();
        System.out.println();
        bike.displayDetails();
        bike.showType();
        System.out.println();
        ev.displayDetails();
        ev.showType();
    }
}
```

7. Output:

```
Campus Parking Vehicle Details:

Vehicle Number: TN01AB1234
Vehicle Type: Car

Vehicle Number: TN02CD5678
Vehicle Type: Two-Wheeler
```